

MONTEVIDEO, MN, USA

WMO: 726553

Lat: 44.969N

Lon: 95.710W

Elev: 315

StdP: 97.60

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04951

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.9	-22.7	-29.9	0.2	-24.2	-27.3	0.3	-22.4	14.1	-12.3	12.2	-11.0	3.1	300	0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	32.4	23.0	31.1	22.6	29.0	21.3	25.2	30.2	24.1	28.8	23.0	27.4	4.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	19.3	28.5	22.5	17.9	27.2	21.4	16.6	26.1	79.1	29.9	73.8	28.8	69.5	28.0	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.7	10.3	8.9	-28.5	35.4	3.5	1.5	-31.0	36.5	-33.0	37.4	-35.0	38.2	-37.5	39.3		
(5)			-28.7	27.3	3.3	1.6	-31.1	28.4	-33.0	29.3	-34.8	30.2	-37.2	31.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.4	-9.8	-7.6	-0.5	7.5	14.7	20.4	22.5	20.8	17.0	9.4	0.9	-6.7		
	DBStd	12.70	7.68	7.32	7.23	5.77	5.04	3.68	3.09	2.97	4.70	5.34	6.42	6.78		
	HDD10.0	2445	613	493	332	113	15	0	0	0	4	78	280	517		
	HDD18.3	4343	872	726	585	328	135	21	4	11	81	282	524	775		
	CDD10.0	1511	0	0	6	37	161	311	386	334	212	58	6	0		
	CDD18.3	368	0	0	0	2	22	81	132	87	39	4	0	0		
	CDH23.3	3449	0	0	2	34	284	760	1222	710	382	54	1	0		
	CDH26.7	1096	0	0	0	8	99	248	414	194	116	14	0	0		
(14) Wind	WSAvg	4.0	4.2	4.3	4.4	4.8	4.5	3.8	3.1	2.9	3.4	4.1	4.2	4.1		
(15) Precipitation	PrecAvg	654	22	23	37	63	85	102	86	79	68	50	33	18		
	PrecMax	992	79	84	150	166	225	266	233	231	154	169	138	85		
	PrecMin	332	0	1	2	4	24	12	3	9	10	0	1	0		
	PrecStd	170	21	23	29	34	47	61	48	46	42	42	30	19		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.8	12.2	22.0	27.3	32.8	34.1	33.9	32.6	32.6	28.7	21.0	10.2		
		MCWB	3.7	8.8	12.9	15.7	18.5	22.5	24.9	24.9	22.0	17.2	12.2	6.6		
	2%	DB	4.1	7.3	16.4	22.9	29.0	31.5	32.3	31.0	29.8	24.0	16.3	6.3		
		MCWB	1.7	4.4	11.2	13.0	17.6	21.6	24.7	24.0	20.8	15.4	9.6	3.0		
	5%	DB	2.4	4.0	12.5	20.1	27.0	29.1	31.0	28.7	27.6	21.1	12.6	3.7		
		MCWB	0.6	2.2	8.8	11.7	16.6	20.2	23.9	22.6	19.4	14.0	8.1	1.5		
	10%	DB	1.0	2.4	8.8	17.3	23.7	27.6	28.9	27.4	25.9	17.7	9.1	2.2		
		MCWB	-0.2	1.0	6.2	10.3	15.0	19.3	22.5	21.1	18.4	12.2	5.7	0.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.2	8.8	14.7	16.9	22.1	25.0	27.4	26.7	24.2	19.3	13.6	7.3		
		MCDB	7.6	11.3	18.5	24.3	28.7	31.0	30.6	30.9	28.9	25.6	18.9	9.0		
	2%	WB	2.2	4.5	11.8	14.6	19.6	23.5	25.6	24.9	22.5	16.9	10.8	3.5		
		MCDB	4.6	7.1	16.0	21.1	25.7	29.0	30.4	29.3	27.5	21.9	14.9	6.0		
	5%	WB	0.8	2.6	8.9	12.7	17.8	22.1	24.5	23.5	21.0	14.9	8.5	1.9		
		MCDB	2.2	4.3	12.8	18.5	23.3	27.4	29.4	27.2	25.6	19.4	12.5	3.6		
	10%	WB	-0.4	1.0	6.2	11.0	16.2	20.6	23.4	22.2	19.5	12.8	6.1	0.5		
		MCDB	0.7	2.5	8.8	16.2	22.0	26.1	28.0	26.1	24.0	17.1	9.4	1.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.0	9.4	9.2	11.0	11.6	10.9	11.0	11.1	12.3	10.8	9.1	8.2		
		MCDBR	10.6	11.2	14.1	17.0	16.2	13.5	12.4	12.9	14.9	16.0	14.0	11.0		
	5% WB	MCWBR	8.8	8.9	9.3	9.1	7.6	6.3	6.7	7.3	7.4	8.6	8.9	8.6		
		MCDWR	9.7	10.5	13.5	14.4	13.0	11.5	11.4	11.4	12.4	13.4	13.2	10.2		
(40) Clear-Sky Solar Irradiance	taub		0.264	0.275	0.307	0.360	0.383	0.394	0.398	0.405	0.365	0.318	0.282	0.262		
		taud	2.452	2.427	2.445	2.340	2.338	2.366	2.338	2.299	2.394	2.513	2.531	2.468		
	Ebn at Noon		870	922	932	900	886	875	866	845	854	859	842	835		
		Edh at Noon	69	85	97	118	122	120	122	102	102	79	64	61		
(44) All-Sky Solar Radiation	RadAvg		1.80	2.86	3.81	4.74	5.46	6.07	6.54	5.41	4.20	2.71	1.76	1.31		
	RadStd		0.21	0.33	0.37	0.44	0.41	0.41	0.23	0.45	0.31	0.37	0.16	0.14		

Historical Trends

		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(m)
(47) Regional (54 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page